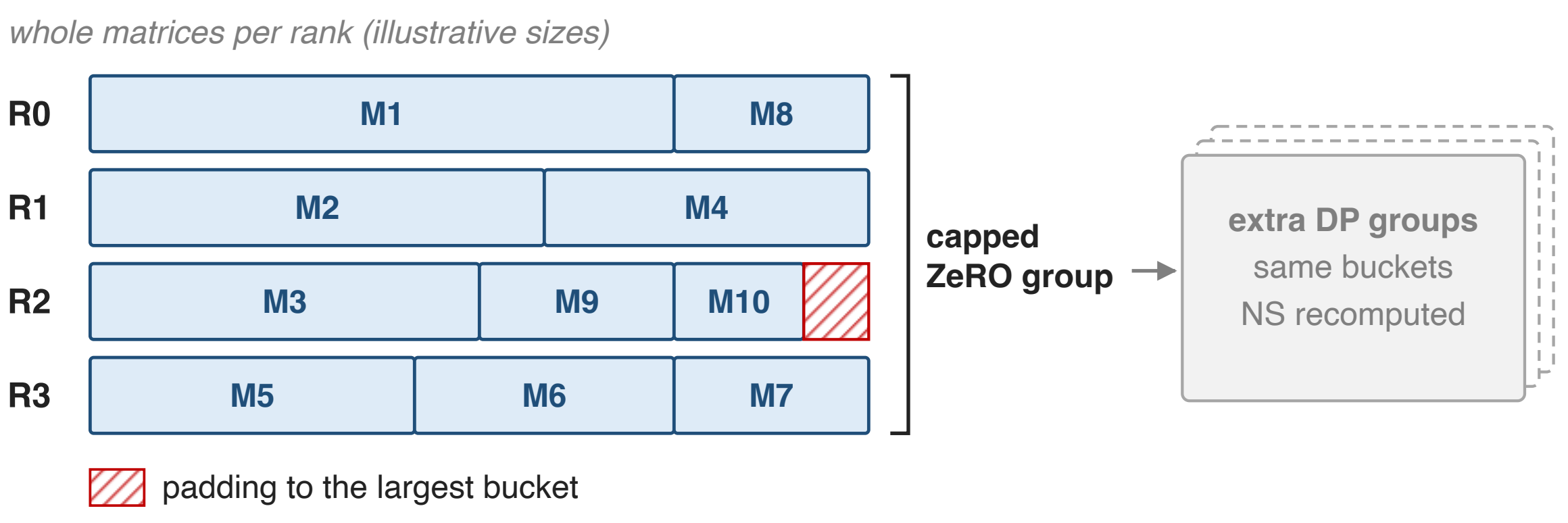

whole matrices per rank (illustrative sizes)
R0
M1
M8
R1
M2
M4
R2
M3
M9
M10
R3
M5
M6
M7
capped
ZeRO group
extra DP groups
same buckets
NS recomputed
padding to the largest bucket